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The Effect of Visualization on Mental Models of Polarization

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Overview

1. Mental models and model understanding
2. The polarization applet
3. Hypotheses and study design
4. Analysis and current status

Mental models and model understanding

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Gestalt and Functionality

Gestalt

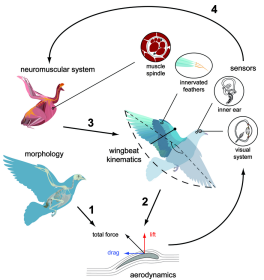
Forms, spatial arrangements, and structures in a learner's mental representation.

Functionality

Relationships and operations used to explain behavior and make predictions.

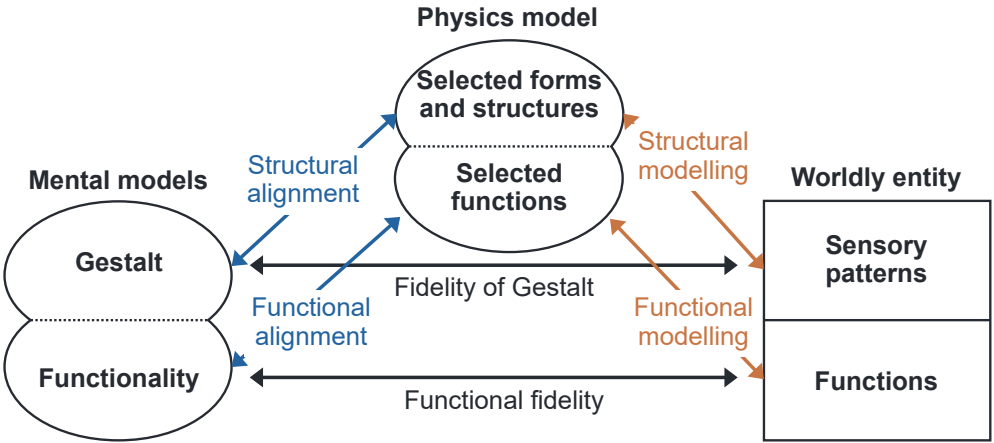


(WikimediaCommons, 2024)



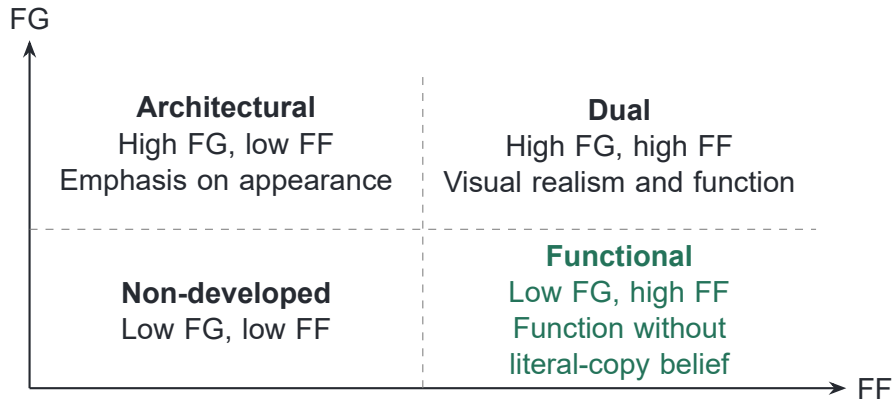
(Altshuler et al., 2015)

Mental models, scientific models, and the world



Collaborative work by Judith Schmid, Malte Ubben, Fabian Hennig, Jonas Bley, and Philipp Bitzenbauer (unpublished, 2026). Black: fidelity; blue: expert alignment; orange: scientific modelling.

Four profiles of model understanding



(Ubben and Bitzenbauer, 2022).

Fidelity and alignment answer different questions

Fidelity: the learner's stance

- What does the learner believe the model is faithful to?
- FG: appearance; FF: function.

Alignment: expert correspondence

- **Structural alignment (StA):** correspondence with expert-selected forms and structures.
- **Functional alignment (FA):** correspondence with expert-selected functions.

High structural alignment can coexist with high FG.

Visualization and the multimedia effect

Multimedia principle

Combining words with relevant pictures can benefit learners more than words alone.

(Mayer, 2005)

DeFT: Design, Functions, Tasks

- **Design:** The number of representations, how information is distributed, the form, the order, and how the translations are highlighted.
- **Functions:** Representations can **complement** each other, **constrain** interpretations, and support the construction of **deeper understanding**.
- **Tasks:** The cognitive tasks required for learning with multiple representations: understanding the representations, their relationships, and selecting accordingly.

(Ainsworth, 2006)

Two pathways for an additional visualization

Representational support

- Complement by allowing different strategies to solve problems.
- Connect notation, geometry, and optical transformations.
- Support expert-aligned explanations and predictions.

Spontaneous reflection

- Encounter different representations of the same state.
- Reconsider literal interpretations.
- Attend to shared functions and limits.

Due to the multimedia principle, expert alignment may improve even when fidelities remain stable.

The polarization applet

Jones vectors and the Poincaré sphere

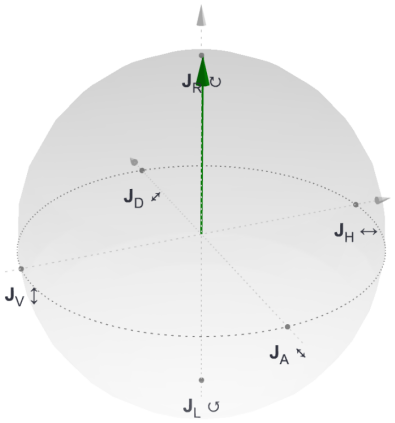
For a normalized, fully polarized state,

$$\mathbf{J} = \begin{pmatrix} a \\ b\, e^{i\varphi} \end{pmatrix}, \qquad a, b \geq 0, \quad a^2 + b^2 = 1.$$

The applet uses

$$\mathbf{S} = \begin{pmatrix} a^2 - b^2 \\ 2ab\cos\varphi \\ 2ab\sin\varphi \end{pmatrix}, \qquad \|\mathbf{S}\| = 1,$$

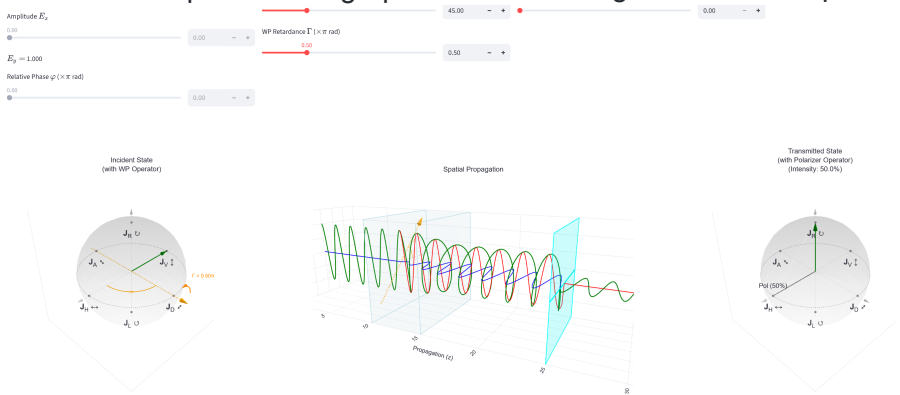
akin to a Qubit on the Bloch sphere
$|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$



The applet

Control:
Jones notation and spatial wave graph.

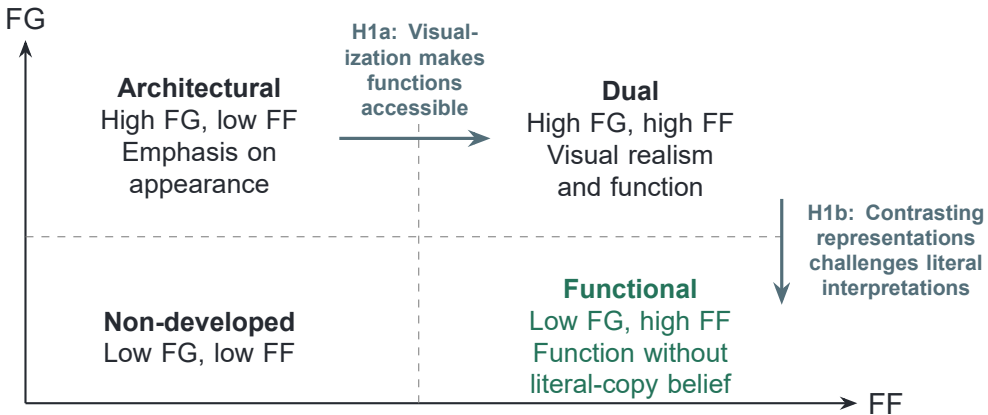
Experimental:
Adding the Poincaré sphere.



+ Introductory explanations, guided parameter changes, hints, and solutions.

Hypotheses and study design

Possible changes in model understanding



Research questions

Main Research Question 1

(How) does the Poincaré sphere visualization change thinking in Fidelity of Gestalt (FG) and/or Functional Fidelity (FF) internal models on polarization?

Main Research Question 2

Does the Poincaré sphere visualization increase alignment with expert views on Gestalt and Function?

Main Research Question 3

(To which extend) are the interactions mediated by cognitive load, spatial reasoning ability, and baseline FG?

Hypotheses

H1a: including the architectural profile

For learners with high FG and low FF, the sphere may make functions accessible: wave plates act as **rotations** of the state, while polarizers perform **projections** of the Jones state.

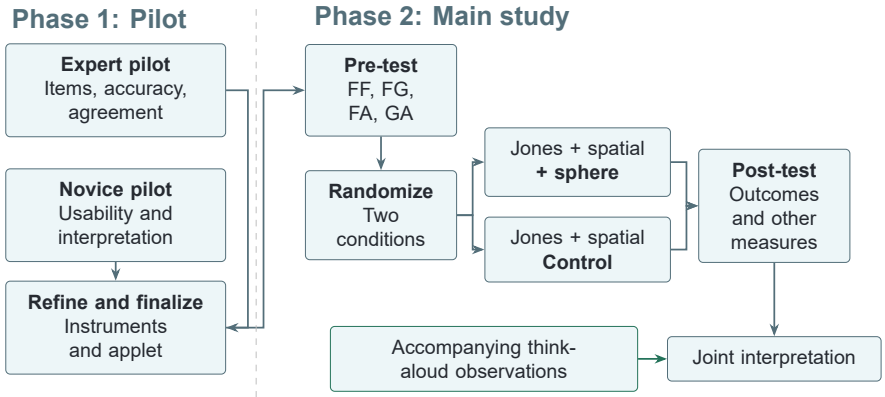
H1b: including the dual profile

For learners with high FG and high FF, contrasting the spatial wave representation with the abstract sphere may challenge literal interpretations and reduce FG while FF stays high.

H2: Multimedia effect

The support with an additional visual representation may increase expert alignment (Rexigel et al., 2024).

Study structure: pilot, then main study



Phase 1: calibrate the instruments and the applet

Expert pilot

- Classify items as FF, FG, both, or neither.
- Judge factual correctness and the usefulness of analogies.
- Evaluate wording and agreement.

Novice pilot

- Identify confusing controls and instructions.
- Examine interpretation of the representations.
- Assess cognitive load (Krieglstein et al., 2023) and technological acceptance (Abdullah and Ward, 2016).

Refine questionnaire items, applet instructions, and interaction design before the main comparison.

Phase 2: randomized pre/post comparison

Pre-test	FF, FG, FA, and GA
Intervention	Random assignment to guided polarization tasks with or without the added sphere
Post-test	FF, FG, FA, GA; cognitive load and spatial reasoning (Young et al., 2025), demographic information

Population: second- and third-year physics / related STEM undergraduates.

Target: $N = 218$ recruited; minimum analyzable sample $N = 196$.

Think-aloud observations: investigating mechanisms

Students verbalize their thoughts while working through the existing applet tasks.

- Observe how students attempt to incorporate the Poincaré sphere into their thinking.
- Trace difficulties in connecting graph, symbolic, and sphere representations.
- Examine spontaneously expressed conflicts between literal interpretations and functional relationships.
- Follow how these conflicts are resolved, maintained, or bypassed during task engagement.

The observations can provide evidence for mechanisms underlying learning and difficulties with mental models.

Analysis and current status

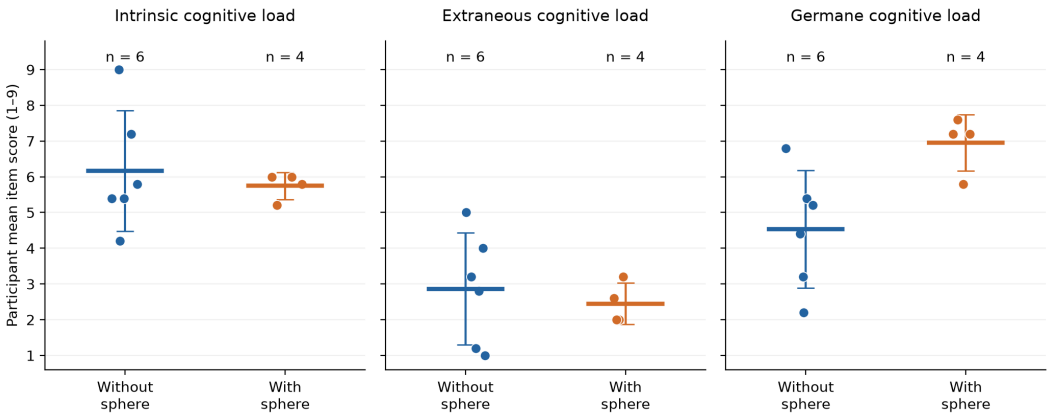
Planned primary analysis: MANCOVA

Joint outcomes

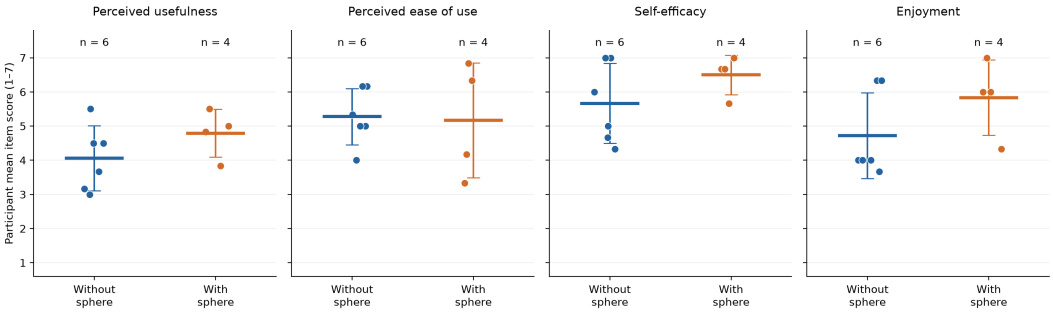
$$\mathbf{Y}_{\text{post}} = \begin{pmatrix} FF_{\text{post}} \\ FG_{\text{post}} \\ FA_{\text{post}} \\ GA_{\text{post}} \end{pmatrix}$$

- **Factor:** condition, with or without the additional Poincaré sphere.
- **Covariates:** baseline outcome scores and spatial reasoning ability to assess MRQ 2.
- **Multivariate test:** assess the adjusted condition effect across the four outcomes.
- **Outcome-specific analyses:** examine the directional predictions of H1a, H1b, and H2.

Initial pilot findings: cognitive load



Initial pilot findings: technological acceptance



Technological Acceptance: Perceived Usefulness and Ease of Use

Perceived Usefulness

- Interactive graphics made the effects of parameter changes immediately visible.
- Hints and direct experimentation supported learning.
- Missing examples and definitions reduced the perceived learning value.

Perceived Ease of Use

- Camera resets, scrolling, and perspective controls disrupted interaction.
- Sliders were sometimes difficult to adjust precisely.
- Participants wanted easier access to instructions, formulas, and previous content.

Technological Acceptance: Self-Efficacy and Enjoyment

Self-Efficacy

- Hints enabled participants to proceed independently.
- Some completed the tasks despite uncertainty about matrices and terminology.
- Task completion did not necessarily indicate conceptual confidence.

Enjoyment

- Participants enjoyed manipulating parameters and observing graphical changes.
- Solving successive tasks created motivation similar to a reward system.
- Complex content and awkward controls reduced enjoyment.

Overall: Interactivity and visualization were valued most. Usability problems and insufficient conceptual guidance were the main barriers. The comments do not reveal a distinct benefit of the Poincaré sphere.

Appendix: The learning sequence

Steps	Topic	Learner activity
1–3	Polarization states	Change amplitudes and relative phase to construct linear and circular states.
4–6	Wave plates	Vary retardance and fast-axis orientation to transform a state.
7	Linear polarizers	Relate transmission-axis orientation to output intensity.
8–9	Combined operators	Coordinate wave plates and polarizers to meet a target.

Nine challenges precede a completion screen and survey return link.

Appendix: measures and protocol decisions

- Cognitive load: Extraneous (5 questions), Intrinsic (5) and Germane (7), 9-point Likert scale (Krieglstein et al., 2023)
- General Extended Technology Acceptance Model for E-Learning (GETAMEL, only in pilot study): (Abdullah and Ward, 2016)
  - Perceived Usefulness
  - Perceived Ease of Use
  - E-Learning Self-Efficacy
  - Perceived Enjoyment
- Spatial reasoning: Measured using the open-source 3D Block Rotation scale from the International Cognitive Ability Resource (ICAR) (Young et al., 2025)

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